

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	DHANUSHREE S	Reg. No.:	192421218
Date:	03/08/2026	Duration:	60 minutes

Code of Conduct

I Dhanushree S-192421218 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Dhanushree S

Annexure A – Architecture Design Assignment

AI-Based Crop Disease Detection and Farmer Advisory Platform

1. Analyze System Requirements

Introduction

Agriculture is one of the most important sectors that contributes significantly to the economy and food production of every country. Healthy crops are essential for ensuring food security and improving the income of farmers. However, crop diseases caused by fungi, bacteria, viruses, and pests reduce agricultural productivity and lead to huge financial losses. Farmers often face difficulties in identifying diseases at an early stage because of limited knowledge and the lack of access to agricultural experts.

Traditional methods of disease identification involve manual inspection or laboratory testing, which are time-consuming, expensive, and not always available in rural areas. Delayed identification of crop diseases can cause severe crop damage, excessive use of pesticides, and reduced crop quality.

With the advancement of Artificial Intelligence, Deep Learning, and Computer Vision, crop diseases can now be detected automatically by analyzing images of infected crop leaves. The AI-Based Crop Disease Detection and Farmer Advisory Platform is designed to help farmers identify crop diseases quickly and accurately. The platform allows farmers to upload crop leaf images through a web or mobile application. The uploaded images are analyzed using a Convolutional Neural Network (CNN), and the system predicts the disease with a confidence score. Based on the prediction, the platform provides treatment recommendations, fertilizer suggestions, pesticide information, and preventive measures. The system also stores prediction history and provides weather-based agricultural advice to improve farming decisions.

The proposed platform aims to improve agricultural productivity, reduce crop losses, and support sustainable farming practices through intelligent technology.

System Objectives

The main objectives of the proposed system are:

-
- Detect crop diseases automatically using Artificial Intelligence and Deep Learning.
 - Provide accurate disease prediction from crop leaf images.
 - Reduce the time required for disease diagnosis.
 - Help farmers identify diseases at an early stage.
 - Recommend suitable pesticides, fertilizers, and treatment methods.
 - Provide preventive measures to reduce future disease occurrence.
 - Improve crop productivity and quality.
 - Reduce financial losses caused by crop diseases.

- Minimize excessive pesticide usage.
 - Maintain disease prediction history for future reference.
 - Provide weather-based farming recommendations.
 - Allow agricultural experts to update disease information and treatment methods.
 - Ensure secure access for farmers, experts, and administrators.
 - Support cloud-based storage and real-time access.
 - Improve overall agricultural decision-making through intelligent advisory services.
-

Functional Requirements

Functional requirements describe the operations performed by the proposed system.

The Farmer Module allows users to register, log in securely, manage their profiles, upload crop leaf images, capture images using a mobile camera, receive disease prediction results, view confidence scores, obtain treatment recommendations, receive fertilizer and pesticide suggestions, access preventive measures, download reports, view prediction history, and receive SMS or email notifications.

The AI Disease Detection Module performs image validation, image preprocessing, feature extraction, disease classification using Convolutional Neural Networks (CNN), confidence score generation, and disease prediction. After processing the uploaded image, the AI module sends the prediction result to the application server.

The Agricultural Expert Module enables experts to log into the system, review disease reports submitted by farmers, update disease information, add new diseases, modify treatment recommendations, and publish agricultural advisory information.

The Administrator Module manages farmers, agricultural experts, disease databases, AI models, reports, notifications, system monitoring, and database backup. The administrator is responsible for maintaining the overall performance and security of the platform.

The Notification Module sends SMS notifications, email alerts, push notifications, disease warnings, weather alerts, and important agricultural announcements to registered farmers.

Non-Functional Requirements

The system should satisfy several quality requirements to ensure reliable operation.

Performance is one of the most important requirements. The platform should process uploaded crop images quickly and provide disease prediction results within two to five seconds. Fast database access and efficient AI processing should ensure smooth system performance.

Scalability is essential because the system should support thousands of farmers simultaneously. It should be capable of processing large numbers of crop images and expanding easily when additional users or crop varieties are introduced.

Security is necessary to protect user information and agricultural data. The platform should implement secure login authentication, password encryption, HTTPS communication, role-based access control, secure API communication, and database security mechanisms.

Reliability ensures that the system consistently provides accurate disease predictions and remains operational even when failures occur. Automatic backup, fault tolerance, and recovery mechanisms should be implemented to improve reliability.

Availability ensures that farmers can access the platform at any time. The application should be available 24 hours a day and 7 days a week through cloud deployment, server redundancy, and automatic failover mechanisms.

Maintainability allows developers to modify, update, and improve the software easily. The modular architecture should support independent updates of the AI model, user interface, and database without affecting the overall system.

Usability is also important. The application should have a simple and user-friendly interface that supports both web and mobile devices. It should be easy to navigate and provide multilingual support to help farmers from different regions.

Quality Attributes

Scalability enables the system to handle increasing numbers of users and image uploads by using cloud infrastructure and load balancing techniques. Additional application servers and AI services can be added without affecting existing users.

Maintainability is achieved through modular software architecture, where each component performs a specific function. This allows developers to update individual modules independently, reducing maintenance effort and improving software quality.

Reliability ensures accurate disease prediction and uninterrupted service. Database replication, cloud backup, fault tolerance, and regular AI model updates improve the overall reliability of the platform.

Availability is achieved through cloud deployment, redundant servers, automatic failover, and continuous monitoring. These mechanisms ensure that the platform remains accessible to users at all times.

Performance is improved by using optimized image preprocessing techniques, efficient CNN models, database indexing, and caching mechanisms. These techniques reduce response time and provide faster disease predictions.

Security is maintained through encrypted passwords, secure user authentication, HTTPS communication, role-based authorization, secure APIs, and regular database backups. These security measures protect user information and prevent unauthorized access.

2. Select a Suitable Software Architecture

The success of the AI-Based Crop Disease Detection and Farmer Advisory Platform depends on selecting an appropriate software architecture. Since the proposed system includes multiple modules such as user management, disease detection, AI prediction, cloud storage, notification services, and external API integration, it requires an architecture that provides flexibility, scalability, maintainability, security, and high performance.

After analyzing the system requirements, Hybrid Software Architecture is selected as the most suitable architecture for the proposed platform. The Hybrid Architecture combines the features of Model–View–Controller (MVC), Client–Server Architecture, and AI Service Architecture. This combination allows different modules of the system to work independently while communicating efficiently with each other.

Model–View–Controller (MVC) Architecture

The MVC architecture divides the application into three independent components: Model, View, and Controller.

The Model is responsible for managing business logic and database operations. It stores and retrieves farmer information, crop details, disease records, treatment recommendations, and prediction history. Any changes made to the data are handled by the Model component.

The View represents the user interface of the application. Farmers, agricultural experts, and administrators interact with the system through the View. It displays login pages, registration forms, image upload screens, prediction results, treatment recommendations, weather information, and reports. The View is designed to be simple, responsive, and easy to use.

The Controller acts as an intermediary between the Model and the View. It receives user requests, validates input data, processes business logic, communicates with the AI prediction service, retrieves information from the database, and sends the appropriate response back to the user interface.

The MVC architecture improves software organization, reduces code complexity, supports code reusability, and simplifies future maintenance and testing.

Client–Server Architecture

The Client–Server Architecture enables communication between users and the centralized server.

The client side consists of the Farmer Mobile Application, Web Application, Agricultural Expert Portal, and Administrator Dashboard. These clients allow users to upload crop images, view disease predictions, manage accounts, and access advisory services.

The server side consists of the Application Server, AI Prediction Server, Database Server, and Notification Server. The server receives requests from clients, processes them, communicates with the AI model and database, and returns the prediction results.

This architecture ensures centralized management of data and resources while supporting multiple users simultaneously. It also improves security by keeping sensitive information on the server instead of storing it on user devices.

AI Service Architecture

The AI Prediction Module is implemented as an independent service because disease detection requires intensive image processing and deep learning computations.

The AI service receives uploaded crop images from the application server. It performs image preprocessing such as resizing, normalization, and noise removal before extracting important image features. The processed image is then analyzed using a trained Convolutional Neural Network (CNN) model to identify the disease. Finally, the AI service returns the disease name and prediction confidence score to the application server.

Separating the AI module from the main application allows developers to retrain or replace the machine learning model without affecting the remaining system components. It also improves scalability because the AI service can be deployed on dedicated high-performance servers.

REST API Communication

Different modules of the proposed platform communicate using REST APIs over HTTPS.

The mobile application communicates with the application server through REST APIs. The application server sends uploaded images to the AI prediction service through REST APIs. The application server also communicates with the weather service, notification service, government agricultural APIs, and cloud storage using secure API requests.

REST API communication provides flexibility, interoperability, and secure data exchange between different services.

Justification for Selecting Hybrid Architecture

The Hybrid Software Architecture is selected because it satisfies both the functional and non-functional requirements of the proposed platform.

The platform performs multiple operations such as user authentication, image upload, disease detection, treatment recommendation, weather advisory, database management, notification services, and report generation. A single architectural style cannot efficiently support all these operations. Therefore, combining multiple architectural styles provides a more efficient solution.

Hybrid Architecture separates different functionalities into independent modules, allowing each module to perform its task without affecting others. The user interface, business logic, AI prediction engine, database management, and external services remain loosely coupled, making the system easier to develop, test, and maintain.

The architecture also supports cloud deployment, enabling the application to serve thousands of farmers simultaneously. If the number of users increases, additional application servers and AI services can be deployed without modifying the existing software.

Security is improved because authentication, authorization, and database management are handled centrally. Communication between modules is protected using HTTPS and secure REST APIs.

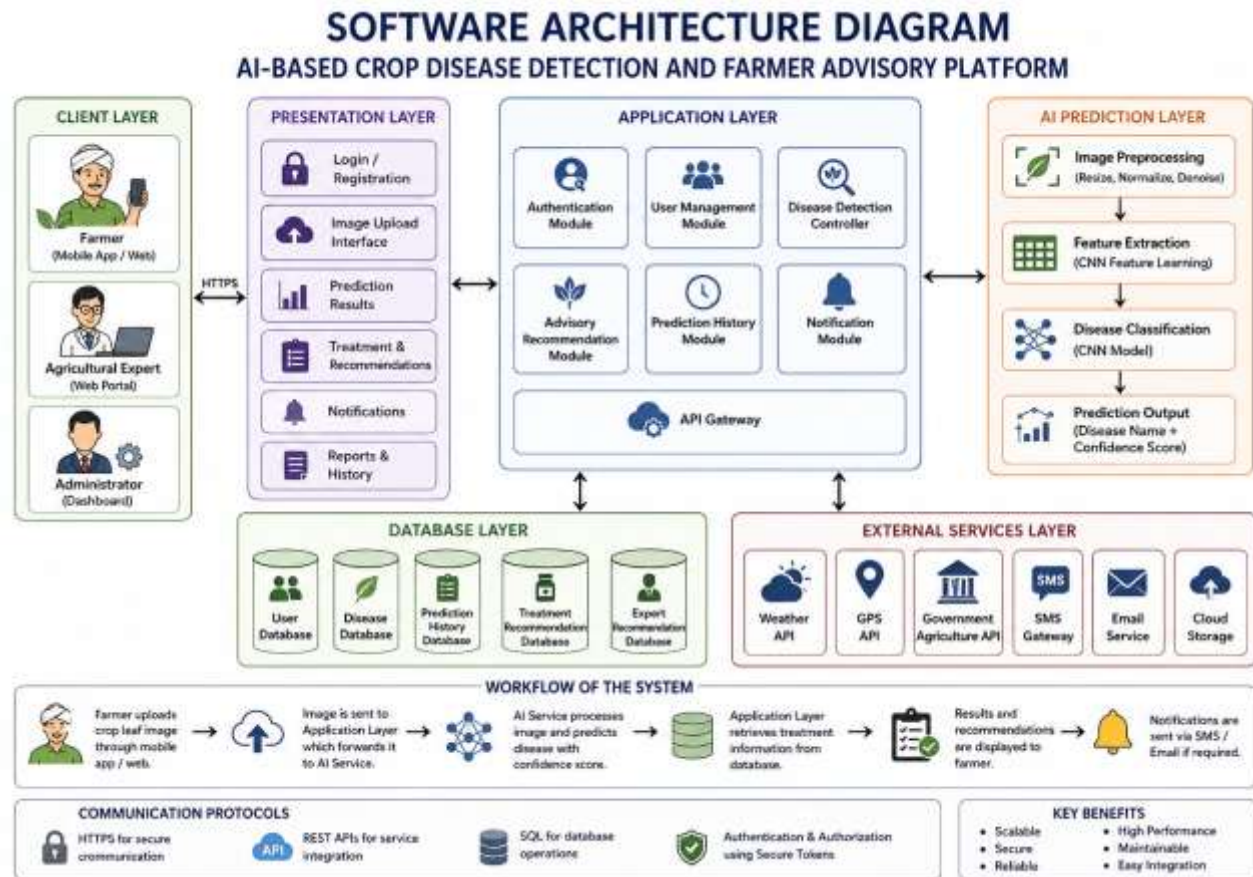
The modular design also makes it easier to integrate new technologies such as IoT sensors, drones, satellite image processing, blockchain, and advanced AI models in the future.

Advantages of Hybrid Architecture

The selected Hybrid Architecture offers several advantages:

-
- Provides a modular software structure.
 - Improves maintainability through separation of components.
 - Supports independent deployment of AI models.
 - Enables cloud-based scalability.
 - Improves system reliability and availability.
 - Supports secure communication using HTTPS and REST APIs.
 - Simplifies testing and debugging.
 - Supports integration with external APIs and cloud services.

- Reduces software maintenance costs.



Explain Components and Their Interactions

4. Explain Components and Their Interactions

The AI-Based Crop Disease Detection and Farmer Advisory Platform is developed using a Hybrid Software Architecture. The system consists of several interconnected components, and each component performs a specific function. Together, these components provide accurate crop disease detection, treatment recommendations, and advisory services to farmers.

4.1 Client Layer

The Client Layer is the interface through which users interact with the system. It includes Farmers, Agricultural Experts, and Administrators.

The Farmer uses the mobile application or web application to register, log in, upload crop leaf images, view disease prediction results, receive treatment recommendations, and access previous prediction history.

The Agricultural Expert logs into the expert portal to review disease reports, update treatment recommendations, add new diseases, and publish farming advisories.

The Administrator manages users, monitors AI model performance, updates databases, generates reports, and controls the overall operation of the system.

The Client Layer sends user requests to the Presentation Layer through secure HTTPS communication.

4.2 Presentation Layer

The Presentation Layer provides the graphical user interface of the application. It includes login pages, registration forms, dashboards, image upload screens, prediction result pages, reports, and notification interfaces.

The main responsibility of this layer is to collect user input, validate basic information, and display outputs received from the Application Layer. It ensures that the system is user-friendly, responsive, and easy to navigate for farmers with different levels of technical knowledge.

4.3 Application Layer

The Application Layer contains the core business logic of the platform. It receives requests from the Presentation Layer and coordinates communication between all other components.

The Authentication Module verifies user credentials and provides secure access based on user roles.

The User Management Module stores and manages user information, including farmer profiles, expert details, and administrator accounts.

The Disease Detection Controller receives uploaded crop images and forwards them to the AI Prediction Service. After receiving the prediction result, it retrieves treatment recommendations from the database and prepares the final response.

The Advisory Recommendation Module provides fertilizer suggestions, pesticide recommendations, preventive measures, and farming advice based on the detected disease.

The Prediction History Module stores all previous disease predictions and allows farmers to review their earlier reports.

The Notification Module sends SMS messages, email alerts, and push notifications regarding disease detection results, weather updates, and agricultural advisories.

4.4 AI Prediction Layer

The AI Prediction Layer is the intelligent component of the system.

After receiving the crop image from the Application Layer, it performs image preprocessing techniques such as resizing, normalization, and noise removal. The processed image is then analyzed using a trained Convolutional Neural Network (CNN). The CNN extracts important image features and classifies the disease. Finally, the AI model generates the disease name along with the prediction confidence score and sends the result back to the Application Layer.

This layer provides fast and accurate disease prediction without requiring human intervention.

4.5 Database Layer

The Database Layer securely stores all application data.

The User Database stores user registration details, login credentials, and profile information.

The Disease Database contains disease names, symptoms, causes, affected crops, and disease descriptions.

The Treatment Recommendation Database stores fertilizer recommendations, pesticide details, treatment procedures, and preventive measures.

The Prediction History Database stores all previous disease predictions and image analysis reports.

The Expert Recommendation Database stores recommendations published by agricultural experts for different crop diseases and seasonal farming practices.

The Database Layer ensures secure storage, fast retrieval, and efficient management of information.

4.6 External Services Layer

The External Services Layer connects the platform with third-party services.

Weather APIs provide weather forecasts and climate information to generate location-based farming recommendations.

GPS APIs identify the user's location and provide region-specific agricultural advice.

Government Agriculture APIs provide information about government schemes, subsidies, and agricultural programs.

SMS Gateway sends text messages containing disease alerts and farming recommendations.

Email Service sends prediction reports and notifications to registered users.

Cloud Storage securely stores uploaded crop images, reports, and backup files.

These external services enhance the functionality of the platform and improve the quality of advisory services provided to farmers.

4.7 Component Interaction

All components communicate with each other using secure communication protocols.

The Farmer first logs into the application and uploads a crop leaf image. The Presentation Layer receives the image and forwards it to the Application Layer. The Application Layer validates the request and sends the image to the AI Prediction Layer through REST APIs.

The AI Prediction Layer preprocesses the image, extracts important features, and predicts the crop disease using the CNN model. The prediction result and confidence score are returned to the Application Layer.

The Application Layer retrieves treatment recommendations, fertilizer suggestions, pesticide information, and preventive measures from the Database Layer. It also checks weather information from the Weather API to provide additional farming advice.

The final prediction report is sent to the Presentation Layer, where it is displayed to the farmer. The system stores the prediction history in the database and sends notifications through SMS or email if required.

Agricultural experts can review disease reports and update treatment information, while administrators monitor the overall system and maintain databases.

Discuss Quality Attributes

5. Discuss Quality Attributes

Quality attributes are essential characteristics that determine the overall efficiency, reliability, and performance of a software system. The proposed Hybrid Software Architecture has been designed to satisfy all major quality attributes required for the AI-Based Crop Disease Detection and Farmer Advisory Platform.

5.1 Scalability

Scalability refers to the ability of the system to handle an increasing number of users, requests, and data without affecting performance.

The proposed platform is deployed on cloud infrastructure, allowing new servers to be added whenever the number of users increases. Since the AI Prediction Service is developed as an independent module, additional AI servers can be deployed to process more crop images simultaneously. The database can also be expanded to store millions of crop images and disease records.

This architecture enables the platform to support thousands of farmers across different geographical locations without performance degradation.

5.2 Security

Security is one of the most important quality attributes because the platform stores user information, crop data, and prediction reports.

The system provides secure user authentication through username and password verification. Passwords are stored using encryption techniques. Role-Based Access Control (RBAC) ensures that farmers, experts, and administrators can only access authorized resources.

Communication between all system components takes place through HTTPS and secure REST APIs. Database backups and regular security monitoring help protect the system against data loss and cyberattacks.

These security mechanisms ensure confidentiality, integrity, and availability of user data.

5.3 Performance

Performance measures how quickly and efficiently the system performs its operations.

The AI model is optimized to analyze crop images and generate disease predictions within a few seconds. Image preprocessing techniques reduce unnecessary processing time, while efficient database indexing speeds up data retrieval.

The use of cloud computing, load balancing, and caching techniques further improves system response time and ensures smooth operation even when many users access the platform simultaneously.

5.4 Reliability

Reliability refers to the ability of the system to operate continuously without failure.

The platform uses cloud-based deployment, automatic backup mechanisms, and redundant servers to ensure uninterrupted service. Fault-tolerant design enables the system to continue functioning even if one server fails.

The AI prediction model is trained using a large dataset of crop images, resulting in accurate disease identification and reducing prediction errors.

These features improve the reliability of the platform and increase user confidence.

5.5 Maintainability

Maintainability refers to the ease of updating, modifying, and improving the software.

The proposed Hybrid Architecture separates the application into independent modules such as the Presentation Layer, Application Layer, AI Prediction Layer, Database Layer, and External Services. Developers can modify one module without affecting the remaining components.

The modular design simplifies debugging, testing, software updates, and AI model improvements, thereby reducing maintenance cost and development time.

5.6 Availability

Availability refers to the percentage of time the system remains operational.

The proposed platform is deployed on cloud servers that provide 24×7 availability. Redundant servers, automatic failover mechanisms, continuous monitoring, and regular backups ensure that the application remains accessible even during hardware failures or maintenance activities.

Farmers can access the platform anytime using mobile devices or web browsers to obtain disease predictions and farming advice.

Summary

The proposed Hybrid Software Architecture successfully satisfies all important quality attributes. It provides excellent scalability for handling large numbers of users, strong security for protecting sensitive data, high performance through optimized AI processing, reliable operation with cloud-based fault tolerance, easy maintenance through modular design, and continuous availability using cloud infrastructure.

Justify Architectural Decisions

6. Justify Architectural Decisions

Selecting the correct software architecture is one of the most important decisions in software development because it directly affects system performance, scalability, security, maintainability, and future expansion. For the AI-Based Crop Disease Detection and Farmer Advisory Platform, a Hybrid Software Architecture has been selected because it effectively combines the strengths of MVC, Client–Server Architecture, and AI Service Architecture.

6.1 Rationale Behind the Architectural Choice

The proposed platform performs multiple functions such as user authentication, image upload, AI-based disease prediction, database management, notification services, and external API integration.

Using only one architectural style would not efficiently support all these requirements. Therefore, the Hybrid Architecture was selected because it provides better modularity, flexibility, and scalability.

MVC Architecture separates the user interface, business logic, and data management, making the software easier to develop and maintain.

Client–Server Architecture allows centralized processing, secure communication, and easy management of multiple users.

The independent AI Prediction Service enables developers to improve or replace the machine learning model without modifying the remaining application.

This combination provides a flexible and future-ready software architecture.

6.2 Trade-offs Considered

Every software architecture has advantages and disadvantages.

The Hybrid Architecture provides high scalability, security, and maintainability but requires slightly more development effort because multiple components must communicate through APIs.

Compared to a simple Layered Architecture, the Hybrid Architecture is more complex to implement. However, the additional complexity is justified because it supports independent AI services, cloud deployment, and future expansion.

Although Microservices Architecture provides excellent scalability, it requires more infrastructure and operational management. For the proposed system, Hybrid Architecture provides sufficient scalability while keeping implementation complexity under control.

Therefore, the selected architecture offers the best balance between performance, flexibility, and development cost.

6.3 Support for Future Enhancements

One of the major advantages of the proposed architecture is its ability to support future technological improvements.

The system can be enhanced by integrating IoT sensors that continuously monitor crop health and environmental conditions.

Drone technology can be incorporated to capture aerial images of large agricultural fields for disease detection.

Satellite image analysis can be integrated to monitor crop health over wide geographical areas.

Voice assistants and multilingual support can be added to help farmers who have limited literacy skills.

Advanced machine learning models with higher prediction accuracy can replace the existing CNN model without changing other system components.

Blockchain technology can be integrated to securely maintain agricultural records and improve transparency.

Government agricultural portals, market price systems, and weather forecasting services can also be connected through REST APIs without redesigning the software architecture.

6.4 Long-Term Maintainability

The modular architecture makes long-term software maintenance easier.

Each module performs an independent function and communicates through well-defined interfaces. Developers can update the AI model, user interface, database, or notification service individually without affecting the remaining modules.

This reduces software maintenance cost, simplifies testing, improves software quality, and extends the lifespan of the application.

The architecture also supports continuous software improvement as new technologies emerge.

Conclusion

The Hybrid Software Architecture is the most suitable choice for the AI-Based Crop Disease Detection and Farmer Advisory Platform. It successfully satisfies all functional and non-functional requirements while providing high scalability, security, performance, reliability, maintainability, and availability.

The modular design allows easy integration of future technologies such as IoT, drones, satellite imagery, cloud computing, and advanced Artificial Intelligence models. This architecture not only meets current system requirements but also provides a strong foundation for future enhancements, making it an ideal solution for modern smart agriculture.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.

Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	Mark
System Requirement Analysis (30%)	/5
Selection of Software Architecture (25%)	/5
Architecture Diagram and Component Design (20%)	/5
Component Interaction and Quality Attribute Analysis (15%)	/5
Architectural Justification and Technical Presentation (10%)	/5
TOTAL	/25

